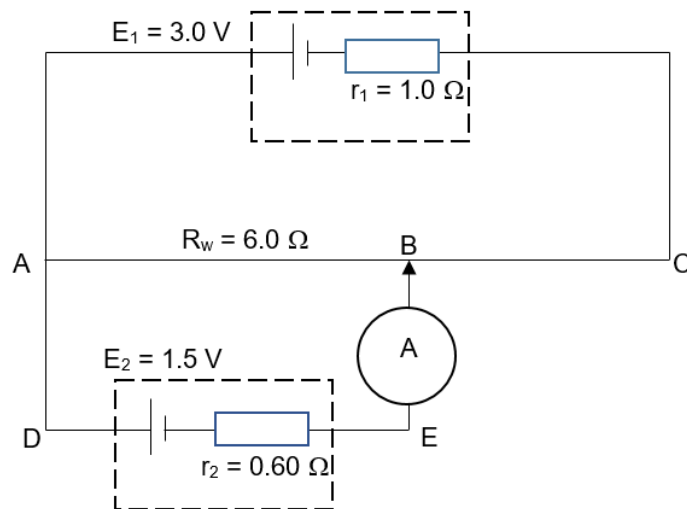




At balance, $V_{AB} = V_{DE}$

Since no current flows through the bottom cell at balance, $V_{DE} = E_2$

$$\text{Also, } \frac{L_{AB}}{L_{AC}} = \frac{V_{AB}}{V_{AC}} = \frac{V_{DE}}{V_{AC}} = \frac{E_2}{V_{AC}}$$

$$\text{Then, } V_{AC} = E_1 \frac{R_w}{R_w + r_1} \quad (\text{apply potential divider principle to the top loop})$$

$$= 3.0 \times \frac{6.0}{6.0 + 1.0}$$

$$= 2.571 \text{ V}$$

$$\text{So balance length, } L_{AB} = L_{AC} \frac{E_2}{V_{AC}}$$

$$= 1.0 \times \frac{1.5}{2.571}$$

$$= 0.58 \text{ m}$$